Sparse support packing for explicit Diophantine consistency equations

Nine positive unknowns; ten unrestricted integer unknowns

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Abstract

The latest multi-target construction in ProveIt distinguishes all 528 monomials of degree at most two in 31 variables, although its 22 source equations use only 71 distinct monomials. We replace complete quadratic separation by the weaker, sufficient requirement that each of these 71 monomials be isolated from every possible contaminating monomial. Explicit certificates reduce the largest weight from 799 to 239 or 220. A second change replaces a cubic variable base by a fixed multiple of a square. Together these changes produce a nine-positive-variable polynomial of degree 23,878,392 with 670 live binary arithmetic operations, and a 651-operation alternative of degree 25,936,016. The compared construction has degree 129,875,980 and costs 788 operations. Applying the sparse weights to the existing unrestricted-integer reduction gives ten-variable polynomials of degree 65,506,504 in 675 operations, or degree 71,151,176 in 662 operations, compared with 237,520,456 in 782 operations.

The new contribution is an explicit, audited simplification of the source encoding. The proof transfers solvability to the same fixed quadratic system used by the repository. Its interpretation as ZFC refutability uses the inherited ordinary mathematical machine/compiler theorem; the ten-integer route also uses Sun's cited reduction. No new end-to-end Lean or Rocq certificate, proof of consistency, or global minimality theorem is claimed. The archive contains the exact finite coefficient payloads, executable polynomial circuits, proofs, and independent checks.

1 Result, meaning, and comparison

Write $\mathrm{Ref}_{\mathrm{ZFC}}$ for the arithmetical assertion that there is a finite formal ZFC proof of a fixed contradiction, and put $\mathrm{Con}(\mathrm{ZFC}) := \neg\,\mathrm{Ref}_{\mathrm{ZFC}}$. We seek a *fixed* polynomial, not a family with an unspecified index or a coefficient chosen according to whether ZFC is consistent. All coefficients in this construction are supplied by finite recipes and materialized binary integers. No recipe tests consistency.

The source snapshot is `124c07c0b77a9e20e84676514d8b1ff78536ea3f`. It includes the wave-6 packages 38–45, although the collection README at that snapshot still mainly describes wave 5. The comparison below uses the actual wave-6 files [2, 3], rather than treating that README as current.

Domain	Construction	Total degree	Binary operations
$\mathbb{Z}^9_{>0}$	Package 45 baseline	129,875,980	788
$\mathbb{Z}^9_{>0}$	New, weights $W = 239$	25,936,016	651
$\mathbb{Z}^9_{>0}$	New, weights $W = 220$	23,878,392	670
$\mathbb{Z}^{10}$	Package 40 baseline	237,520,456	782
$\mathbb{Z}^{10}$	New, weights $W = 239$	71,151,176	662
$\mathbb{Z}^{10}$	New, weights $W = 220$	65,506,504	675

Both new weight choices improve both displayed measures over their respective baselines. Within each new pair, the degree and operation objectives differ: $W = 220$ has lower degree, while $W = 239$ is

cheaper to evaluate. The latter has a more favorable addition-chain and expression sharing structure; a smaller exponent does not automatically imply fewer gates.

For nine positive variables, the degree improvement is a factor of approximately 5.44 at $W = 220$, and the operation reduction is approximately 17.4% at $W = 239$. For ten unrestricted integers, the corresponding figures are approximately 3.63 and 15.3%. These are substantial reductions in the low-variable part of the repository's frontier. They do not beat the separate 67-operation higher-variable equation or the quartic alternatives on their preferred measures.

Arithmetic convention. Every live binary $+$, $-$, or $\times$ costs one operation, including multiplication by a fixed numeral. Reusing a register is free. Fixed powers are emitted as multiplication gates. A displayed abbreviation is a polynomial expression, not a new existential variable. Degree is total degree in the quantified variables; fixed integer coefficients have degree zero. The JSON circuits are complete division-free descriptions of ordinary polynomials in $\mathbb{Z}[X_1, \ldots, X_n]$.

Exact scope of the theorem. The unconditional new transfer result is

$$\exists \boldsymbol{x} \in \mathbb{Z}_{>0}^{9}\ F_{9,W}(\boldsymbol{x}) = 0 \quad \Longleftrightarrow \quad \exists \boldsymbol{u} \in \mathbb{Z}_{>0}^{31}\ \bigwedge_{j=1}^{22} r_j(\boldsymbol{u}) = 0. \tag{1}$$

The right side is the *same fixed system*, with the same complete four-coefficient tuple, as package 31. Composing (1) with its inherited machine/compiler interpretation gives the asserted consistency equivalence, by ordinary mathematics formalizable in ZFC:

$$\mathrm{ZFC} \ \vdash\ \Big(\neg\exists \boldsymbol{x} \in \mathbb{Z}_{>0}^{9}\ F_{9,W}(\boldsymbol{x}) = 0\Big) \longleftrightarrow \mathrm{Con(ZFC)}. \tag{2}$$

The analogous composition applies to the ten-integer construction. Equation (2) inherits the ordinary compiler proof as a mathematical dependency; this note does not turn the repository's open formal compiler and object-language proof-export obligations into completed kernel certificates. Section 9 records that distinction explicitly.

2 The fixed quadratic source

The 31 positive source coordinates, in the order used by the weight vectors, are

$$J, F, \alpha, z, f, h, i, o, j, s, w, \tau, \eta, \zeta, y, \gamma,$$
$$N, E, P, c, a, X, D, C_2, L, F_2, V, g, I, B_{\mathrm{aux}}, \Delta.$$

The four fixed ports are (β, K_0, M_C, M_F) from package 31's `wang943-mixed-121x8801` instance. The native base is $\mathcal{B} = 2^{1{,}064{,}921}$, and $\beta = \mathcal{B} - 1$. Here M_F is the *shifted* high mask. Mixing this port with an unshifted mask, or taking ports from different compiler tuples, would define a different problem.

Define only for writing the source equations

$$q = \beta J + 1, \quad k = \eta + \zeta, \quad H = 4a + 3, \quad C = q - F - \alpha,$$
$$U = L + 1, \quad Y = a - E, \quad Q = c - \eta, \quad T = P - k, \quad R = jc - V.$$

The ordered equations are

$$\begin{aligned}
&U = q^2, && N = qU, && X = wq,\\
&Y = sN, && E = XY, && Q = kY,\\
&T = EQ, && \tau^2 = PT + 1, && D = X + ac + \gamma H,\\
&\Delta = a^2 + H, && C_2 = c^2, && D^2 = \Delta C_2 + 1,\\
&k = R + 1 + hE,
\end{aligned}$$

$$\begin{aligned}
R &= (q - F)(N - q) - CL + (M_C + qM_F)J,\\
(K_0 + w)C &= F + z\beta J,\\
F_2 = f^2, &\qquad V = of - c, \qquad g = y^2 - V^2,\\
I = iC_2, &\qquad B_{\text{aux}} = I^2, \qquad B_{\text{aux}} = \Delta(F_2 - 1),\\
y^2 - B_{\text{aux}} g &= 1.
\end{aligned}$$

All 22 left-minus-right residuals are quadratic. This is the exact 60-operation comparison system in `quartic31_system60.json`, whose SHA-256 is

`ac2a67782d344d1e74fd93bc3f09fae0097eaed1829fd8194ae235ae84763e48.`

The archive preserves that source file and the complete coefficient payloads. The port bit lengths are respectively

$$1,064,921, \quad 1,070,004, \quad 1,064,921, \quad 1,064,922.$$

Rebuilding the tuple from the literal machine transitions and 943 tiles reproduces the pinned hashes. Its four uncompressed payloads total 533,099 bytes; their compressed forms total 10,485 bytes. These are the native inputs, not the coefficient list of a fully expanded final polynomial.

Set each positive source coordinate to $x_i + 1$ and regard $x_i \geq 0$. Reorder the rows as

$$1, 2, 3, 4, 6, 7, \ldots, 22, 5.$$

Call the reordered shifted residuals $p_1, \ldots, p_{22}$. The last row has constant coefficient 1. This has two uses: the tuple $\boldsymbol{x} = 0$ is not a common zero, and the top coefficient of the encoding polynomial below is positive and independent of the ports. Independent sparse polynomial replay gives 136 row terms, with 71 distinct monomials in their union: one constant, 31 linear monomials, and 39 quadratic monomials.

3 Only used monomials need to be separated

Let

$$\mathcal{A} = \bigcup_j \operatorname{supp}(p_j), \qquad \mathcal{Q} = \{\alpha \in \mathbb{N}^{31} : |\alpha| \leq 2\}.$$

Thus $|\mathcal{A}| = 71$ and $|\mathcal{Q}| = \binom{33}{2} = 528$. The condition actually needed is

$$\alpha \in \mathcal{A},\ \delta \in \mathcal{Q}, \quad \alpha \cdot \boldsymbol{w} = \delta \cdot \boldsymbol{w} \quad \Longrightarrow \quad \alpha = \delta. \tag{3}$$

There is no requirement that two monomials outside $\mathcal{A}$ have different weights. Their collisions never select a nonzero source coefficient. Full B_2 separation is sufficient but stronger than (3).

Here are the two delivered vectors, in the coordinate order of Section 2:

$$\begin{aligned}\boldsymbol{w}_{220} = (&24, 89, 95, 146, 205, 155, 147, 197, 203, 78, 29,\\ &148, 47, 208, 25, 217, 129, 169, 30, 215, 220, 219, 135,\\ &166, 27, 133, 32, 200, 43, 204, 152),\\ \boldsymbol{w}_{239} = (&237, 231, 223, 221, 233, 193, 195, 215, 120, 190, 226,\\ &142, 125, 151, 148, 169, 214, 134, 140, 121, 126, 172, 128,\\ &196, 238, 191, 239, 192, 197, 184, 205).\end{aligned}$$

An exact finite certificate checks (3) against all 528 expansion monomials for every used monomial. The optimizer is not trusted: any method may propose the vector, and a short independent verifier checks the property. No minimality of either vector is asserted.

For either vector put

$$W = \max_i w_i, \quad d = 2W + 1, \quad h_j = jd - 1, \quad H_* = 22d - 1, \quad K_* = H_* + 2W.$$

If $p_j(\boldsymbol{x}) = \sum_\alpha p_{j,\alpha} \boldsymbol{x}^\alpha$, define the fixed integers

$$c_{j,\alpha} = (2 - |\alpha|)! \left(\prod_i \alpha_i! \right) p_{j,\alpha}$$

and the univariate polynomial

$$\mathcal{D}(Z) = \sum_{j,\alpha} c_{j,\alpha} Z^{h_j - \alpha \cdot \boldsymbol{w}}. \tag{4}$$

All exponents in (4) are nonnegative fixed integers. The four-port expressions for the 136 coefficients are given exactly in the data files. The first target is $2W$, the lowest position that uniformly accommodates every quadratic term. This one-place shift saves three operations in each delivered nine-variable evaluation schedule.

Lemma 3.1 (Sparse extraction)**.** *For every integer tuple $\boldsymbol{x}$ and every j,*

$$[Z^{h_j}]\ \mathcal{D}(Z) \left(1 + \sum_i x_i Z^{w_i} \right)^2 = 2p_j(\boldsymbol{x}). \tag{5}$$

Proof. A source term in row k and an expansion monomial δ reach h_j exactly when

$$(k - j)d = \alpha \cdot \boldsymbol{w} - \delta \cdot \boldsymbol{w}.$$

The right side has absolute value at most $2W$, so $k = j$. Property (3) then forces $\alpha = \delta$. The coefficient in the squared expansion is $2!/((2 - |\alpha|)! \prod_i \alpha_i!)$; multiplying by the coefficient in (4) gives exactly $2p_{j,\alpha}$. □

The intervals $[h_j - 2W, h_j]$ for the different source rows are disjoint. Within each row, (3) makes its used exponents distinct. Therefore every coefficient of $\mathcal{D}$ is one $c_{j,\alpha}$, not a hidden sum of several source coefficients. Its highest coefficient, at H_*, is 2. Moreover, if $|c_{j,\alpha}| < M$, every coefficient of the product in (5) has absolute value at most

$$M \left(1 + \sum_i |x_i| \right)^2. \tag{6}$$

This bound is unchanged by collisions among unused expansion monomials: one sums their multinomial coefficients, not the number of distinct weights.

4 A fixed multiple of a square replaces the cubic base

The supplied coefficient recipes and the port bit lengths give the explicit bound

$$M = 2^{2,134,927}, \qquad \lambda = 4096M = 2^{2,134,939}. \tag{7}$$

The verifier computes the bound from the integer recipes, rather than accepting the exponent as an unexplained parameter. Work temporarily with nine coordinates

$$\varepsilon, \phi, n \geq 0, \qquad g, h, s, w, i, j \geq 1. \tag{8}$$

Set

$$b = \varepsilon + 2, \qquad B = \lambda b^2. \tag{9}$$

In the old construction b had a large additive floor and $B = b^3$. The reason for that cubic was a coefficient bound proportional to Mb^2. Since M is fixed, placing it in a fixed multiplicative coefficient suffices: $1024Mb^2 = B/4$. This removes one unit of variable degree from every packed power without introducing a variable. The fixed value λ is exactly package 45's former additive `base_floor`, now used as a multiplicative `base_scale`. The other 16 fixed leaves are unchanged. Thus this improvement does not introduce a larger fixed coefficient or enlarge the fixed-leaf payload: the 17 compressed nine-variable payloads total 64,286 bytes.

Define the following polynomial registers:

$$\begin{aligned}
\ell &= \sum_{r=1}^{31} B^{w_r}, & D_c &= \mathcal{D}(B),\\
N_0 &= B^{W+1}, & N_1 &= 4B^{K_*+1}, \qquad N = N_0 N_1,\\
M_c &= N_0 - 1 - (b-1)\ell, & T_* &= \sum_{r=1}^{22} B^{h_r},\\
T_c &= M_c + (B-2)T_* N_0,\\
J_c &= 2(1+g)^2 D_c + B\sum_{r=0}^{K_*} B^r,\\
S_c &= g + J_c N_0,\\
R &= (N-1)\big((S_c + T_c + 1)N + T_c + 1\big),\\
G_c &= 32bB^W.
\end{aligned}$$

The subscript c distinguishes coding registers from similarly named quantities in the later Pell construction. Every sum is finite with a fixed upper endpoint. Geometric sums are computed polynomially; a fraction such as $(B^{k+1}-1)/(B-1)$ is not an instruction in the circuit.

Lemma 4.1 (Uniform outer bounds)**.** *On every assignment (8), $D_c, M_c, T_c, J_c, S_c > 0$, and $R, N \geq 8$, $b \leq R, N$. If also $g < 4bB^W$, then*

$$0 < g < N_0, \qquad 0 < J_c < N_1, \qquad 0 < S_c, T_c < N. \tag{10}$$

Proof. Since $B - 1 > M$ and the top coefficient of $\mathcal{D}$ is 2,

$$D_c > 2B^{H_*} - M\sum_{r<H_*} B^r > B^{H_*} > 0, \qquad D_c < \frac{MB^{H_*+1}}{B-1}.$$

Also $\ell < (B^{W+1}-1)/(B-1)$ and $b-1 < B-1$, so $M_c > 0$. The other unconditional assertions follow immediately from the displayed positive formulas. On the bounded-g branch, $4b < B$ and $1+g \le 4bB^W$, so

$$2(1+g)^2 D_c < \frac{32Mb^2}{B-1} B^{K_*+1}.$$

Here $32Mb^2 = B/128$. Adding the geometric-mask term yields $0 < J_c < 4B^{K_*+1} = N_1$. The target-mask bound $(B-2)T_* < B^{H_*+1}$ gives $T_c < N$, while $g < N_0$ and $J_c < N_1$ give $S_c < N$. □

5 Why the carry code imposes every source equation

Assume first that b is a power of two and $g < 4bB^W$. Then B, N_0, N_1, N are powers of two. For $N = 2^r$ and $0 \le S, T < N$, the identity

$$(N-1)((S+T+1)N+T+1) = (S+T)N^2 + (N-1-S)N + (N-1-T) \tag{11}$$

has nonoverlapping binary blocks. Thus

$$\operatorname{pop}(R) = 2r + \operatorname{pop}(S_c + T_c) - \operatorname{pop}(S_c) - \operatorname{pop}(T_c).$$

Legendre's factorial identity gives $v_2\binom{2R}{R} = \operatorname{pop}(R)$. Consequently

$$N^2 \mid \binom{2R}{R} \quad \Longleftrightarrow \quad \text{adding } S_c \text{ and } T_c \text{ produces no binary carry.} \tag{12}$$

The low-block condition is that g and M_c have disjoint 1-bits. The base-B digit of M_c is $B-b$ at each allowed w_i and $B-1$ at every other position through W. Therefore this condition is equivalent to

$$g = \sum_i x_i B^{w_i}, \qquad 0 \le x_i < b. \tag{13}$$

The bound $g < N_0$ excludes any additional higher digit.

For such g, let u_r be the coefficients of $\mathcal{D}(Z)(1+\sum_i x_i Z^{w_i})^2$. Its degree is at most K_*. By (6),

$$|u_r| \le M(32b)^2 = B/4 < B/2.$$

Hence $a_r = u_r + B/2$ satisfies $0 < a_r < B$ and

$$J_c = 2\sum_{r=0}^{K_*} a_r B^r, \qquad (B-2)T_* = 2\sum_{j=1}^{22} (B/2-1)B^{h_j}.$$

Dividing both operands by 2 is a common binary shift. The target digit $B/2-1$ uses all the low bits of its cell. Disjointness forces the positive digit a_{h_j} to be exactly $B/2$. By (5), this is precisely $p_j(\boldsymbol{x}) = 0$, separately for each of the 22 rows. There is no inference from a sum of residuals to their separate vanishing.

Conversely, given a common nonnegative zero $\boldsymbol{x}$ of the source, choose any sufficiently large power of two $b > \max_i x_i$, with $b \ge 2$. Equation (9) supplies a power-of-two B. Set $\varepsilon = b-2$ and define g by (13). Since the all-zero tuple is not a source zero, $g > 0$. Moreover $g < (b-1)\sum_{r \le W} B^r < bB^W$. Both no-carry conditions hold, and therefore

$$N^2 \mid \binom{2R}{R}, \quad b \text{ is a power of two}, \quad 32g < G_c. \tag{14}$$

The weaker inequality $8g < G_c$ supplied by the soundness direction of the ratio theorem gives exactly the bounded-g hypothesis used above. This ordering avoids circular use of the digit bound.

6 The nine-variable polynomial, explicitly

Use the coding registers just defined, and the remaining coordinates $h, s, w, i, j > 0$, $\phi, n \geq 0$. Define

$$\begin{aligned}
U &= N^2 w, & Y &= N^2 s, & M_0 &= RY,\\
P &= 2M_0^2 U, & K &= R + 1 + h(P-1), & A &= M_0(U+1),\\
B_1 &= 2R+1, & W_1 &= bw, & C &= B_1 + W_1 + \phi,\\
J &= 4A - 5, & Q &= 3W_1 C - 2(W_1^2 - 1), & &\\
D &= (A^2-1)C^2 + 1, & E &= iJDC^2, & F &= (A^2-1)E^2 + 1,\\
G &= A + F(CD + 1 - A), & H &= B_1 + jC, & I &= (G^2-1)H^2 + 1,\\
a_0 &= (P^2-1)K^2 + 1, & a_1 &= DFI, & &\\
u &= JF, & v &= QF + J(H - C), & &\\
m &= K^2 G_c - 4(C - KY)^2 G_c - 8gK^2. & & & &
\end{aligned}$$

The numerical Jones ratio theorem states that, under the unconditional bounds already proved, the four conditions

$$a_0, a_1 \text{ are integer squares}, \quad u \mid v, \quad m > 0 \tag{15}$$

imply the central-binomial divisibility, the power-of-two condition on b, and $8g < G_c$. Conversely (14) supplies witnesses satisfying (15). The precise numerical theorem is `ratio_sound` and `ratio_complete` in namespace `Jones1982.Nine` of the pinned `NineRatio.lean`; it does not fix the old positional weights [6, 5]. Thus changing the source coding does not silently change a hypothesis of the ratio theorem.

To express (15) with the one remaining coordinate n, put

$$V_0 = PK, \qquad V_1 = (AC)(AE)(GH), \qquad M_d = Q + H,$$

$$\Omega = um, \qquad \Theta = u(n + M_d) + v - \Omega(2M_d + V_0 V_1).$$

The equation is

$$\boxed{F_{9,W} = \left[\Theta^2 - \Omega^2(a_0 + V_0^2 a_1)\right]^2 - 4\Omega^4 V_0^2 a_0 a_1 = 0.} \tag{16}$$

Together with Sections 2–4, this is a finite definition of the polynomial. It contains no variable exponents, divisions, binomial coefficients, square-root functions, or extra quantifiers. Those objects occur only in its correctness proof.

Lemma 6.1 (Finalizer)**.** *For the preceding outer parameters, an $n \geq 0$ satisfying (16) exists if and only if (15) holds.*

Proof. All of $P, K, A, C, J, D, E, F, G, H, I$ are positive, and the formulas give

$$0 < a_0 < V_0^2, \qquad 0 < a_1 < V_1^2, \qquad 0 < v/u < M_d - 1. \tag{17}$$

For example $V_0^2 - a_0 = K^2 - 1 > 0$; the three inequalities $(AC)^2 > D$, $(AE)^2 > F$, and $(GH)^2 > I$ give the second bound. Also $H - C = B_1 + (j-1)C > 0$, and $v/u = Q/J + (H-C)/F < Q + H - 1$.

If $m = 0$, then $\Omega = 0$ and $\Theta/u = n + M_d + v/u > 0$, so (16) cannot hold. If $m \neq 0$, the four-factor identity for (16) gives

$$\Theta/\Omega = r_0 + V_0 r_1, \qquad r_0^2 = a_0, \quad r_1^2 = a_1.$$

The roots are real. The rational number on the left cannot vanish: otherwise $|r_1| = |r_0|/V_0 < 1$, contradicting $a_1 \geq 1$. Writing it as $x \neq 0$ gives $2xr_0 = x^2 + a_0 - V_0^2 a_1$, so r_0 and then r_1 are rational. A rational square root of an integer is integral.

By (17), $|r_0| \leq V_0 - 1$ and $|r_1| \leq V_1 - 1$. Hence

$$Z = 2M_d + V_0V_1 + r_0 + V_0r_1 \geq 2M_d + 1.$$

The vanishing factor says $n + M_d + v/u = mZ$. Its left side is positive, so $m > 0$. Its right side is an integer, so $u \mid v$. Conversely, if (15) holds, choose integral roots and set $n = mZ - M_d - v/u$. The same bounds make this an integer at least 2, and the corresponding factor vanishes. □

Composition with the carry-code lemma proves (1). To obtain exactly nine *positive* input variables, replace ε, ϕ, n by their positive input coordinates minus 1. To obtain nine *nonnegative* input variables instead, shift g, h, s, w, i, j upward by 1. These are different supplied circuits. The positive circuit must not be interpreted as an unrestricted-integer equation.

7 Exact degree and why the leading cancellation matters

Let

$$\zeta = H_* + 3W + 2 = 47W + 23, \qquad t = \deg N = 2\zeta = 94W + 46.$$

On the diagonal where each quantified variable equals an indeterminate T, the leading terms of N and R are $4\lambda^{\zeta}T^t$ and $16\lambda^{3\zeta}T^{3t}$. The high geometric term in J_c dominates its sparse-coefficient term, and the S_cN^2 term dominates R.

The following table records the degrees needed for an analytic certificate. All are also full multivariate upper bounds.

Register	Degree	Register	Degree
P	$12t + 3$	K	$12t + 4$
A	$7t + 2$	C	$3t$
D	$20t + 4$	E	$33t + 7$
F	$80t + 18$	G	$103t + 22$
H	$3t + 1$	I	$212t + 46$
a_0	$48t + 14$	a_1	$312t + 68$
u	$87t + 20$	m	$28t + 2W + 11$
Ω	$115t + 2W + 31$	V_0	$24t + 7$
V_1	$156t + 34$	M_d	$3t + 2$

Naive degree propagation through (16) overestimates the result, because the highest terms of V_1^2 and a_1 cancel. Resolve this by the exact identity

$$\Delta_V := V_1^2 - a_1 = (C^2 - 1)(AE)^2(GH)^2 + D(E^2 - 1)(GH)^2 + DF(H^2 - 1).$$

Its first two summands have degree $298t + 64$ and each has leading coefficient $2^{518}\lambda^{298\zeta}$; their sum does not cancel. With $E_0 = u(n + M_d) + v - 2\Omega M_d$, the inner bracket in (16) equals

$$E_0^2 - 2E_0\Omega V_0V_1 + \Omega^2V_0^2\Delta_V - \Omega^2 a_0.$$

The third term uniquely dominates. It has degree $576t+4W+140$ and leading coefficient $2^{1025}\lambda^{576\zeta+2W}$. The final term subtracted in (16) has strictly lower degree. Therefore

$$\boxed{\deg F_{9,W} = 1152t + 8W + 280 = 108296W + 53272.} \tag{18}$$

The exact nonzero diagonal leading coefficient is

$$2^{2050}\lambda^{1152\zeta+4W} = 2^{2050}\lambda^{54148W+26496}. \tag{19}$$

Thus the upper bound is attained for the actual fixed coefficients, not merely for a generic formal coefficient tuple. The separately exported cancellation-resolved circuit checks the same identity. Modular leading term checks provide additional independent transcription evidence; (19) is the mathematical reason the leading term is nonzero.

The effects of the changes can be separated:

Source encoding	Maximum weight	Nine-variable degree
Baseline, cubic base	799	129,875,980
Only change to scaled square base	799	86,584,080
Only use sparse weights	220	35,820,904
Sparse weights and scaled square base	220	23,880,696
Also shift all targets down by one	220	23,878,392

The coefficient heights of expanded polynomials are enormous in all these representations. We compare exact circuits and fixed coefficient payloads, not an ungenerated dense monomial list.

8 Ten unrestricted integer variables

The same sparse support certificate can be inserted into package 40's quadratic source and its existing ten-integer tail. That baseline already uses a fixed multiple of a square as its packing base, and already uses the quintic divisibility merger

$$(r^2 + 23S^2)(r^3 - rS^2 - S^3), \qquad r = T + SY.$$

Neither device is a new contribution of this note on the ten-integer side. The change is the source's finite exponent vector and the corresponding regenerated powers, masks, coefficient placement, and evaluation schedule. The Sun-dependent bridge and finalizer are retained [2, 7].

The complete formulas and literal division-free circuits are included in Appendix A and the data files. Its ten inputs range independently over *all* integers. The source-isolation proof is precisely Section 3; its carry bounds have the same fixed quadratic source degree and coefficient bound. Changing the positional weights does not change the numerical hypotheses of the inherited tail.

The exact degree is

$$\boxed{\deg F_{10,W} = 297088W + 147144.} \tag{20}$$

For $W = 799$, this recovers the pinned baseline value 237,520,456; for $W = 239$ and 220 it gives 71,151,176 and 65,506,504. The supplied degree verifier follows the actual fixed-coefficient circuit and checks the required nonzero leaders. The unrestricted-integer claim retains the same reliance on Sun's 2026 preprint as the existing ten-integer construction. The nine-variable positive construction is independent of that preprint.

9 Proof dependencies and formalization status

There are three logically different achievements here.

1. The new finite-data assertions are completely checkable: source residual replay, isolation of every used monomial, coefficient binding, gate liveness, operation counts, and degree certificates.
2. The new ordinary proof establishes the sparse extraction, scaled base bounds, carry coding, and transfer to the fixed source system. Its unbounded steps are mathematical proofs, not extrapolations from testing small possible Diophantine solutions.
3. The interpretation of that fixed source as ZFC refutability is inherited from package 31's compiler proof and the calibrated K215 endpoint. These are cited mathematical dependencies, not consequences of finite circuit tests.

The active research status at the pinned commit explicitly distinguishes an unconditional Lean theorem about a classically selected nine-variable polynomial from the explicit native-window circuits, and lists the explicit compiler formalization and object-language ZFC derivation as open tasks [4]. The polynomial constructed here is specified by computable coefficient data; it is not that classically selected polynomial. Likewise, a local Lean proof of equality of two zero sets does not by itself identify either set with ZFC refutations.

Our ordinary ZFC-provability claim uses the imported compiler theorem in exactly the same way as its source construction. A reader requiring an end-to-end machine-checked certificate should regard that dependency as a remaining verification obligation. We neither assume Con(ZFC) nor claim to decide it. No inference to an internal ZFC proof is made merely from the soundness of Lean as a metatheory.

10 Reproduction, audit design, and further reductions

The archive is self-contained for its numerical and algebraic checks. Python's standard library suffices. The release driver checks the parent data and the materialized ports, rebuilds the selected circuits, and validates both weight choices. The source verification covers every used monomial against every possible quadratic expansion monomial, and also replays all 136 row terms. The degree verification uses an explicit cancellation identity; it is not an interpolation at a handful of points. Proof supplements and the immutable source ledger identify which external theorems are imported.

The polynomial can be evaluated modulo any integer directly from its JSON register list. Exact evaluation at even small integer assignments can create impractically large integers, so modular formula checks are the useful reproducible transcription tests. They do not search for solutions or establish consistency.

Why this change has room to generalize. For a fixed degree-d source, one need only isolate source support monomials from the degree-d expansion monomials that can contaminate their coefficient targets. A full B_d code need not be optimal for that task. Here this distinction is enough to remove most of the irrelevant pair-separation constraints. The method is not specific to K215; the source's sparse support is the optimization input.

Most useful next research directions. The next finite optimization is to search the actual isolation constraints with exact solvers or improved local search and to optimize operation count together with maximum weight. Merely reducing W can worsen the addition-chain cost, as the two

delivered vectors show. A second possibility is to move the 22 target positions independently instead of spacing them uniformly by $2W+1$; each proposed packing must still be checked against cross-row contamination and coefficient overlap. A third is to change the quadratic source coordinates so that its used support is easier to isolate. These are further research questions, not improvements asserted by this release.

The most consequential verification task is separate: formalize the explicit native compiler and connect its literal coefficient payloads to the already checked ZFC refutation endpoint. The smaller source encoding here can then be added as a compact arithmetic theorem with a finite isolation certificate.

A Complete ten-integer definition and transfer proof

This supplement specifies the second delivered polynomial without any implicit auxiliary unknowns. Its ten independent unknowns are

$$(f, g, h, k, l, w, x, y, z, m) \in \mathbb{Z}^{10}.$$

Every other symbol in this section abbreviates an integer polynomial in these ten variables. Symbols used for the nine-variable construction are reset locally. The parent polynomials p_j, their fixed weighted coefficients $c_{j,\alpha}$, and either one of the two weight vectors are exactly those in Sections 2–3. In particular,

$$W = \max_i w_i, \quad h_j = j(2W+1) - 1, \quad H_* = 44W + 21, \quad K_* = 46W + 21,$$

and $\mathcal{D}$ is the fixed univariate polynomial in (4). These are fixed exponents and fixed coefficients, not further unknowns.

A.1 The two-input packed source

Set the fixed integer $\beta_{10} = 2^{2140030}$ and define

$$b = 1 + 3f, \qquad B_{10} = \beta_{10} b^2, \qquad \ell = \sum_{i=1}^{31} B_{10}^{w_i},$$

$$N_0 = B_{10}^{W+1}, \qquad N_1 = 4B_{10}^{K_*+1}, \qquad N = N_0 N_1,$$

$$M_c = N_0 - 1 - (b-1)\ell, \qquad Q_c = (B_{10} - 2)\sum_{j=1}^{22} B_{10}^{h_j}, \qquad T_c = M_c + Q_c N_0,$$

$$J_c = 2(1+g)^2 \mathcal{D}(B_{10}) + B_{10} \sum_{q=0}^{K_*} B_{10}^q, \qquad S_c = g + J_c N_0.$$

Finally put

$$X = (N-1)\big((S_c + T_c + 1)N + T_c + 1\big), \qquad Y = N^2, \qquad C_b = bB_{10}^W.$$

Thus $N = 4B_{10}^{47W+23}$ and $C_b = 2^{2140030W} b^{2W+1}$. All sums have fixed finite limits. The released circuit evaluates them by shared powers and geometric-sum identities.

A.2 The retained Lucas and norm registers

Define, in the displayed order,

$$\begin{aligned}
L &= lY, & U &= 2LX, & V &= 4gwY, & W_b &= bw,\\
A &= U(V+1), & B_1 &= 2X+1, & C &= B_1 + (A-2)h,\\
P_n &= U^2V, & K &= X+1+k(P_n-2),\\
S &= 2A-5, & T &= 3W_bC - 2(W_b^2-1),\\
R_2 &= (P_n^2-4)K^2+4.
\end{aligned}$$

The margin and Lucas registers are

$$\begin{aligned}
\Delta_b &= 8C_b^3gK^2 - g^2(32(C-KL)^2C_b^3 + g^2K^2),\\
O &= f^2l^2\Delta_b, & \xi &= zx,\\
D &= (A^2-4)C^2+4, & E &= C^2D\xi,\\
F &= 4(A^2-4)E^2+1, & G^* &= 2(A+2)(A-2)^2E^2-1,\\
\Omega_{10} &= 16E, & H^* &= C + (B_1-C)F + \Omega_{10}Fy,\\
I^* &= ((G^*)^2-1)(H^*)^2+1.
\end{aligned}$$

The homogeneous quintic merger and the three radicands are

$$\begin{aligned}
\Sigma &= \xi S^5, & r &= T + Sy,\\
\Theta_{10} &= (r^3 - rS^2 - S^3)(r^2+23S^2),\\
r_0 &= b(1+(3O-4)b^2z^2), & r_1 &= R_2, & r_2 &= DFI^*,\\
v_1 &= r_0, & v_2 &= v_1(P_n^2+K^2+3),\\
Z_n &= \Sigma m - \Theta_{10},\\
a_0 &= \Sigma^2 r_0, & a_1 &= \Sigma^2 v_1^2 r_1, & a_2 &= \Sigma^2 v_2^2 r_2.
\end{aligned}$$

Set $V_n = Z_n^2 + a_2 - a_0 - a_1$. The complete equation is

$$\boxed{F_{10} := (V_n^2 + 4Z_n^2a_2 - 4a_0a_1)^2 - 16V_n^2Z_n^2a_2 = 0.} \tag{A.1}$$

After the definitions are substituted, this is a single polynomial over $\mathbb{Z}$ in precisely the ten displayed integer unknowns. The files `sparse220_ten_integer.json` and `sparse239_ten_integer.json` specify the corresponding expanded binary arithmetic schedules and bind every nonliteral leaf to its exact materialized coefficient.

A.3 The precise inherited source interface

Here is the interface of package 40's source theorem used in the composition [2]. Let H_0 bound the ordinary coefficients of the shifted parent rows. One may take

$$H_0 = 8(1+\beta+K_0+M_C+M_F)^2, \qquad \beta_{10} > 4\cdot 33^2 H_0.$$

The displayed inequality follows from the fixed port bit lengths; the exact coefficient payloads also verify it directly. Then:

(i) If the p_j have a common zero in $\mathbb{N}^{31}$, then for every sufficiently large positive integer f for which $b = 1 + 3f$ is both a square and a power of two, there is a g with

$$b \le g < C_b, \qquad Y \mid \binom{2X}{X}.$$

(ii) If $f \neq 0$, b is a square and a power of two, $0 \le g < 2C_b$, and $Y \mid \binom{2X}{X}$, then the p_j have a common zero in $\mathbb{N}^{31}$.

(iii) Already under $f \neq 0$, b a square, and $0 \le g < 2C_b$, without the power-of-two assumption, one has

$$0 \le S_c, T_c < N, \quad 3 \mid X, \quad X > 3b, \quad Y > \max\{b, 256\}.$$

The new support-isolation property proves the exact same coefficient identities and coefficient bound used in that theorem. Distinct encoded positions, positive leading coefficient 2, target spacing, and nonnegative exponents have all been retained. The proof of (iii) bounds J_c for every integer in the broad permitted interval before any digit recovery. For (ii), the lower mask recovers $g = \sum_i n_i B_{10}^{w_i}$ with $0 \le n_i < b$; the upper mask tests all 22 centered coefficients, and the binary carry identity gives the binomial divisibility. For (i), choose arbitrarily large $b = 4^q > \max_i n_i$ and $f = (b-1)/3$, then use the same word. The parent zero is nonzero because its last row has constant coefficient 1; thus $g \ge B_{10} \ge b$. These observations prove that the source interface transfers to the two supplied vectors.

Sun's retained argument uses this interface, $b = 1 + 3f$, and $C_b = 2^c b^d$ with c even and d odd. Here $c = 2140030W$ is even and $d = 2W + 1$ is odd. Thus the new source has all the required congruence and growth hypotheses. The use is of the parametric proofs of Propositions 6.1–6.2 and Theorem 4.1 of Sun's 2026 paper, as composed and documented in packages 31 and 40 [7, 1, 2].

A.4 Why the unrestricted norm has exactly the required zeros

The retained bridge has two relevant consequences. A common parent zero supplies positive witnesses with $A > 2$ even, $B_1 > 0$ odd, b square, $b \le g < C_b$, $C = u_{B_1}(A)$, R_2 square, $S \mid T$, and $O > 0$. Conversely, for the formulas above,

$$b \text{ square}, \quad O > 0, \quad \xi \neq 0, \quad DFI^* \text{ square}, \quad R_2 \text{ square}, \quad S \mid T \tag{S1}$$

imply a parent zero. The reverse proof first obtains the source's broad ranges from $O > 0$; it then recovers the Lucas value and the power-of-two base. The full interpolation assertion says that, when $C = u_{B_1}(A)$ under the positive hypotheses, for any positive d one may choose $\xi > 0$ divisible by d such that *every* residue class modulo ξ contains a y making DFI^* a square.

For completeness, the finalizer enforces exactly (S1) as follows. The polynomial $\pi(t) = (t^3 - t - 1)(t^2 + 23)$ has no rational zero and has a root modulo every positive integer. Its cubic discriminant is -23: at an odd prime other than 23, either the cubic has a simple root or its irreducible Frobenius 3-cycle makes -23 a square; at 23 the cubic has simple root 3; at 2 one uses $-23 \equiv 1 \pmod 8$. Lifting and CRT finish the local assertion. As $S = 2A - 5$ is always odd and nonzero, $\Theta_{10} \neq 0$ always. Moreover, $\Sigma \mid \Theta_{10}$ implies $S \mid T$: if $\nu_p(S) > \nu_p(T)$, the two homogeneous factors have valuations $3\nu_p(T)$ and $2\nu_p(T)$, contradicting $S^5 \mid \Theta_{10}$. Conversely, from $S \mid T$ a root of π modulo $|\xi|$ prescribes a residue class for y giving $\Sigma \mid \Theta_{10}$.

For $z \neq 0$, the factors b and $1 + (3O - 4)b^2 z^2$ are coprime, and the second cannot vanish. Therefore r_0 square makes b a signed square; $b \equiv 1 \pmod 3$ excludes the negative sign. Its second factor is then a square, which forces $O > 0$. Conversely positive bridge witnesses have $O \ge g \ge b \ge 4$. Pell's equation

for the positive nonsquare $3O - 4 \equiv 2 \pmod 3$, after taking a unit power whose second coordinate is divisible by b, supplies $z > 0$ making r_0 square.

Finally, for $z \neq 0$ one has $|r_0| > 1$ and

$$\sqrt{|r_1|} \le P_n^2 + K^2 + 2, \qquad |v_1| > \sqrt{|r_0|}, \qquad |v_2| > \sqrt{|r_0|} + |v_1|\sqrt{|r_1|}.$$

These inequalities ensure that a rational signed sum $\pm\sqrt{r_0} \pm v_1\sqrt{r_1} \pm v_2\sqrt{r_2}$ requires all three radicands to be integer squares. Indeed, a multiquadratic-field automorphism changing a root changes the highest-index changed term by more than the combined absolute values of all earlier changes. The argument also covers negative radicands and dependent square classes.

Equation (A.1) is identically the product of its eight signed-root factors. If $\Sigma = 0$, it equals $\Theta_{10}^8 \neq 0$; hence any zero has $\xi = zx \neq 0$. Dividing a vanishing factor by Σ and using separation forces the three squares and then $\Sigma \mid \Theta_{10}$. The preceding mergers give (S1), so the reverse bridge supplies a parent zero. In the other direction, choose the positive bridge witnesses, then the Pell value $z > 0$, then an interpolation modulus ξ divisible by z, then a y in the quintic-prescribed residue class. Put $x = \xi/z$. All radicands are squares and $\Sigma \mid \Theta_{10}$, so an integer m makes one norm factor vanish. This proves

$$(\exists \mathbf{t} \in \mathbb{Z}^{10})\ F_{10}(\mathbf{t}) = 0 \quad \Longleftrightarrow \quad (\exists \boldsymbol{n} \in \mathbb{N}^{31})\ \bigwedge_j p_j(\boldsymbol{n}) = 0.$$

Together with the inherited fixed-compiler theorem, classical negation gives the asserted equivalence with Con(ZFC) in ZFC. The argument retains the cited ordinary proof dependencies and makes no additional claim of end-to-end proof-assistant certification.

A.5 Exact degree and supplied evaluators

Writing $p = 47W + 23$, the source degrees are $d_X = 6p$, $d_Y = 4p$, and $d_C = 2W + 1$. The retained tail gives

$$\begin{aligned}
\deg Z_n &= 5d_X + 10d_Y + 30,\\
\deg a_0 &= 14d_X + 28d_Y + 3d_C + 57,\\
\deg a_1 &= 26d_X + 48d_Y + 6d_C + 98,\\
\deg a_2 &= 114d_X + 224d_Y + 6d_C + 440.
\end{aligned}$$

The last strictly dominates the degrees of a_0, a_1, Z_n^2. Thus the top homogeneous part of F_{10} is the fourth power of the top homogeneous part of a_2, and

$$\deg F_{10} = 297088W + 147144.$$

The $W = 220$ evaluator uses 675 operations and has exact degree 65,506,504; the $W = 239$ evaluator uses 662 operations and has exact degree 71,151,176. Independent direct leading-coefficient propagation on these two exported DAGs gives nonzero residues 489390939 and 862326449 modulo 1,000,000,007, respectively, certifying attainment of the propagated upper bounds.

B A secondary improvement: minimum norms at degree 44

The optional `minima/` supplement reduces the historical 15-positive-variable, degree-44 sum of squares from 78 binary operations to 76, or to 75 using its inherited signed index/transport recovery. The source is package 12's `ordinary_gap_system61.json`, copied byte for byte from commit `5b0c9ae9fdae83db563f9d2b61ea0fb1f4a801e6`. The supplement's `SOURCE_PINS.json` records full paths and blob identities. Both new polynomials bind the complete package-31 coefficient tuple through local binary payloads.

Take fifteen positive variables

$$J, F, \alpha, z, f, h, i, T, s, w, \tau, \eta, \zeta, e, \gamma.$$

Use the fixed ports (β, K_0, M_C, M_F) of Section 2. Define polynomial abbreviations

$$\begin{gathered}
q = \beta J + 1, \quad C = q - F - \alpha, \quad X = wq, \quad Y = sq^3, \quad E = XY, \\
k = \eta + \zeta, \quad c = kY + \eta, \quad a = E + Y, \quad H = 4a + 3, \quad \Delta = a^2 + H, \\
D = X + ac + \gamma H, \qquad R = ((q-1)(q-F) + \alpha)(q^2 - 1) + (M_C + qM_F)J, \\
V = c(Tf - 1) - Rf^2, \quad U = \Delta(f^2 - 1), \quad I = ic^2, \quad W_s = I^2 - U, \\
N_0 = \tau^2 - XY^2(XY^2 + 1)k^2, \quad N_1 = D^2 - \Delta c^2, \\
N_a = U\big(V^2 - (V + e)^2\big) + (V + e)^2, \qquad N_k = k - hE - R.
\end{gathered}$$

For original transport coordinate z put $N_t = (K_0 + w)C + (q - F) - z(q - 1)$. The cheaper equation is

$$\boxed{P_{75} = N_0^2 + N_1^2 + N_a^2 + W_s^2 + (N_k N_t - 1)^2 - 3 = 0.}$$

For the shifted coordinate $z = z_{\text{old}} - 1$, put $R_t = (K_0 + w)C - F - z(q - 1)$. The cleaner local alternative is

$$\boxed{P_{76} = N_0^2 + N_1^2 + N_a^2 + W_s^2 + (N_k - 1)^2 + R_t^2 - 3 = 0.}$$

Why subtracting three is sound. On every positive assignment,

$$N_0^2 \geq 1, \qquad N_1^2 \geq 1, \qquad N_a^2 + W_s^2 \geq 1.$$

Their equality cases are respectively $N_0 = 1$, $N_1 = 1$, and $(N_a, W_s) = (1, 0)$. For the first norm, $n_0 = XY^2 \geq 2$ makes $n_0(n_0 + 1)$ nonsquare, so the norm cannot vanish. A norm -1 solution $\xi^2 - n_0(n_0 + 1)\nu^2 = -1$ would descend under multiplication by $2n_0 + 1 - 2\sqrt{n_0(n_0 + 1)}$: its new second coordinate is $(2n_0 + 1)\nu - 2\xi$, strictly between 0 and ν. For the main norm, $\Delta = (a + 2)^2 - 1$ is nonsquare; its residue 0 or 3 modulo four also excludes norm -1.

If $U > 0$, then $U \geq 2$. A vanishing auxiliary norm would give $UV^2 = (U - 1)(V + e)^2$; since $e > 0$, the two roots cannot both vanish. It would follow that $U(U - 1)$ is a square, contrary to its position between consecutive squares. Thus $N_a \neq 0$. If $U = 0$, then $W_s = I^2 \geq 81$, so the block is larger than 1. Equality in the block therefore forces $W_s = 0$ and $N_a = \pm 1$. Now $U = I^2$, and reduction modulo four makes N_a congruent to V^2 or $(V + e)^2$, excluding -1.

Consequently $P_{76} = 0$ is tuplewise equivalent to the six historical comparisons $N_0 = N_1 = N_a = N_k = 1$, $W_s = R_t = 0$. For P_{75} the same argument gives $N_0 = N_1 = N_a = 1$, $W_s = 0$, and $N_k N_t = 1$. These give a zero of the ordinary product with $y = V + e$, which may be signed. The current package-31 theorem in `upstream/ORDINARY_BLANK_PROOF.md` applies directly on its nine-positive-coordinate domain and

recovers $N_k = N_t = 1$ for the authentic native tuple. Package 12's canonical gap theorem supplies $e > 0$ for completeness. This is the ordinary strong equation $I^2 = \Delta(f^2 - 1)$; it does not assert coordinatewise equivalence to the normalized strong equation. The semantic compiler and ZFC interpretation remain inherited dependencies.

Exact finite certificate. The 76-operation schedule uses 40 multiplications, 21 additions and 15 subtractions; the 75-operation schedule uses 40, 20 and 15. Every gate and every supplied input is live. The cancellation

$$N_a = V^2 - (U-1)e(2V+e)$$

gives the required degree-22 auxiliary bound. Setting all fifteen variables equal to t, exact symbolic propagation yields degree 44 with leading coefficient

$$16\beta^{28} + 256\beta^{24} + 16\beta^{22} > 0.$$

Thus the total-degree upper bound is attained for the actual coefficients. The standard-library verifier authenticates all source and payload bytes, rebuilds both circuits, proves the displayed identity by exact ring expansion, and checks this leading coefficient. Its calculation uses no interpolation. This secondary saving is three operations relative to the historical degree-44 equation; it does not replace the main nine-variable improvement.

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